Adaptive Noise Cancellation for Audio Signals

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Presentation Outline

1. **Introduction**
   ANC concepts & industry context
2. **Noises and its types**
   defining noises & its types in modern era
3. **Literature Overview**
   DSP, ML & hardware references
4. **Research Gap**
   Triple-gap in existing ANC systems
5. **Problem Statement**
   Objectives & design goals
6. **Proposed Methodology**
   Architecture, algorithm & design
7. **Conclusion**
   Findings, roadmap & future work

Understanding Adaptive Noise Cancellation

The Fundamental Principle

- Sound is a pressure wave. Noise cancellation works by generating an "anti-phase" wave (180° out of phase).
- When the peaks of the noise meet the troughs of the anti-noise, they cancel out. This is known as **Destructive Interference**.
- The goal is to create a "Zone of Silence" around the listener's ear or microphone.

The Growing Necessity

- **Health:** Long-term exposure to ambient noise (above 85 dB) leads to permanent hearing loss and increased stress.
- **Clarity:** In our modern remote-work era, background noise is the primary barrier to effective digital communication.
- **Fidelity:** High-end audio enthusiasts require silence to hear the full dynamic range of their music without turning the volume to dangerous levels.

The Evolution of ANC (1930s – 2000s)

1. The Analog Era (The Beginning)

- **1933:** Paul Lueg first patented the concept of "Theory of Noise Feed-forward."
- **Method:** Used purely analog vacuum tubes and basic phase-shifting circuits.
- **Limitation:** It was bulky, consumed massive power, and could only handle very simple, low-frequency hums.

2. The Digital Signal Processing (DSP) Era

- **1980s – 2010s:** The rise of microprocessors allowed for the "LMS" (Least Mean Squares) algorithm.
- **Method:** The system began "learning" noise in real-time using mathematical feedback loops.
- **Success:** This led to the first commercial ANC headphones. It was excellent at stopping constant sounds like airplane engines.

The Modern Shift to AI and Spatial Logic (2020s – Present)

3. The Machine Learning Revolution

- **2020s – Present:** The industry shifted from mathematical feedback loops to Deep Learning.
- **The Predictive Advantage:** Traditional math (FxLMS) is "reactive." AI is "predictive," recognizing noise patterns before they fully reach the ear.
- **Handling Complexity:** AI can distinguish between human speech and "Babble noise," allowing it to preserve the voice while deleting background chaos.

4. Current Industry Implementations

- **Proprietary Hardware:** Industry leaders (Apple, Sony) use custom silicon. Highly efficient, but restricted to expensive, closed-ecosystem wireless devices.
- **Cloud & OS-Level Software:** Solutions like Zoom or Krisp use software. Effective, but limited by OS scheduling, causing audio lag and high power drain.
- **The Gap:** There is a critical lack of high-speed, hardware-agnostic platforms that perform AI-based noise cancellation for standard wired equipment.

Noise Profiling: Defining Noise Types

The Origin of Acoustic Noise

- **Definition:** Noise is any unwanted acoustic energy that physically enters the microphone pathway, overlapping and corrupting the target audio signal.
- **How it Enters:** It originates from environmental sources, reflects off physical surfaces, and is captured by the sensor alongside the user's voice.

1. Continuous Machine Hum (e.g., HVAC & Fans)

- **Characteristics:** 50 Hz – 500 Hz; features sharp, constant tonal peaks.
- **Data:** Power levels range from 50 dB to 65 dB SPL.
- **Modeling Approach:** Modeled mathematically as a sum of harmonically related sinusoids with constant amplitudes and phases.
- **Engineering Note:** Easiest to cancel using traditional phase-inversion.

Noise Profiling: Deep Rumble & Unpredictable Sounds

2. **Deep Engine Rumble (e.g., Aviation & Transit)**
   - **Characteristics:** 50 Hz – 1000 Hz; dense energy spread across the spectrum.
   - **Data:** Sustained high power levels (80 dB – 85 dB SPL).
   - **Modeling Approach:** Modeled as "colored noise" (like pink or brown noise) by passing white Gaussian noise through a low-pass shaping filter.
3. **Unpredictable City Sounds (e.g., Traffic & Alarms)**
   - **Characteristics:** 100 Hz – 2000 Hz; highly unpredictable amplitude spikes.
   - **Data:** Fluctuating between 70 dB – 80 dB SPL.
   - **Modeling Approach:** Modeled using time-varying statistical models like Hidden Markov Models (HMM) or transient impulse functions.

Noise Profiling: Human Noise & Prominence Analysis

4. Human Background Chatter (e.g., Crowd Babble)

- **Characteristics:** 300 Hz – 3400 Hz; overlaps exactly with the primary user's speech.
- **Modeling Approach:** Modeled as a superposition of multiple, independent speech streams that have been randomly delayed and attenuated.

The Prominence Factor in Modern Environments

Among these classifications, **Human Background Chatter** and **Unpredictable City Sounds** are the most prominent and disruptive in the modern remote-communication era. Because these noises occupy the exact same frequency band (300 Hz - 3.4 kHz) and mimic the acoustic patterns of the target user, traditional frequency-based DSP filters cannot remove them without muting the user's voice as well.

This exact prominence is what necessitates the shift toward **Spatial Beamforming** (tracking physical location) and **AI Deep Learning** (recognizing non-linear patterns) to successfully isolate the target signal.

Literature Overview: Signal Processing Foundations

Reference	Algorithm Focus	Contribution / Relevance
1. Boll (1979)	**Spectral Subtraction**	The foundational algorithm for estimating the magnitude of the noise spectrum and subtracting it from noisy speech.
2. Berouti (1979)	**Over-subtraction**	Introduced the spectral over-subtraction method to mitigate the "musical noise" artifacts left by standard spectral subtraction.
3. Widrow (1975)	**Adaptive Filters (LMS)**	The pioneering framework for using a reference noise signal to adaptively minimize the mean square error (MSE).
4. Ephraim & Malah (1984)	**MMSE Estimator**	Developed the Minimum Mean Square Error amplitude estimator, providing superior speech quality compared to basic subtraction.

Literature Overview: Machine Learning & Deep Learning

Reference	Algorithm Focus	Contribution / Relevance
5. Park et al. (2023)	**CNN-Assisted ANC**	Replaces traditional math with a Convolutional Neural Network (CNN) to classify noises. **Forms our core filtering logic.**
6. Valin et al. (2018)	**RNNoise (Hybrid)**	Demonstrated a hybrid approach combining traditional DSP with Deep Learning for real-time, full-band speech enhancement.
7. Defossez et al. (2020)	**Demucs (Waveform)**	Real-time speech enhancement operating directly in the raw audio waveform domain rather than using spectrograms.
8. Braun et al. (2020)	**RNN Deep Noise Sup.**	Focused on recurrent neural networks (RNNs) for handling sudden, non-stationary environmental noises with high accuracy.

Literature Overview: Hardware Implementation

Reference	Hardware Focus	Contribution / Relevance
9. Shi et al. (2020)	**FPGA FxLMS**	Implemented the standard multichannel FxLMS on FPGA using a multiple-parallel-branch folding architecture.
10. Kuo et al. (2022)	**FPGA CNN ANC**	Mapped a feedforward CNN directly onto FPGA logic gates for in-ear headphones. **Our architectural hardware guide.**
11. Wang et al. (2024)	**Embedded Beam-forming**	Implemented spatial beamforming and TDOA on embedded systems. **Provides our system with directional awareness.**
12. Gao et al. (2021)	**Low-Latency NN Logic**	Developed ultra-low latency hardware architectures specifically for deploying neural network-based ANC systems at the edge.

Research Gap: Identifying Current Limitations

Our analysis identifies a critical "Triple-Gap" in existing ANC technology:

- **1. The Mathematical Gap (Stationarity)**
  - **Limitation:** Standard FxLMS is designed for "stationary" noise (steady hums).
  - **The Gap:** Fails to suppress "transient" sounds (dog barks, keyboard clicks) because mathematical adaptation cannot track rapid amplitude spikes.
- **2. The Latency-Intelligence Trade-off**
  - **Limitation:** AI-based filters (e.g., Krisp) are smart but run on General CPUs/GPUs.
  - **The Gap:** High processing overhead causes 20-50ms lag. Research lacks high-intelligence filters operating at **sub-10ms hardware speeds**.
- **3. The Spatial Blind-Spot**
  - **Limitation:** Most systems use 1 or 2 mics, focusing purely on frequency spectra.
  - **The Gap:** Frequency-only filters cannot isolate voice from background "Babble" (crowd noise). There is a lack of real-time **3-mic spatial logic** for location-based isolation.

Problem Statement and Research Objective

The Problem Statement

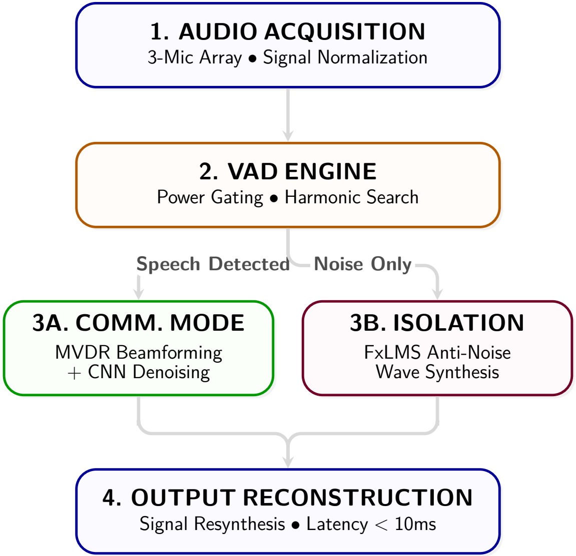
Current acoustic noise cancellation systems force a critical compromise between speed and intelligence. Traditional mathematical filters are fast enough for real-time communication but fail completely when unpredictable, everyday noises—like background chatter or sudden traffic—overlap with the user's voice. Conversely, modern Artificial Intelligence can easily untangle these complex sounds, but it introduces heavy computational delays that disrupt live, two-way conversations. The core challenge this project addresses is the **Intelligence-Speed Paradox**: *How can we harness the superior pattern recognition capabilities of deep learning while maintaining the ultra-low latency required for seamless, real-world audio communication?*

Research Objectives

To resolve this paradox, this research aims to:

- **Hybrid Framework:** Develop an architecture merging deterministic signal processing speed with AI-driven pattern recognition.
- **Spatial Discrimination:** Implement multi-mic logic to isolate sound sources based on physical Arrival Angles (TDOA).
- **Latency Optimization:** Achieve total processing delay < 10ms, staying well below the human perceptibility threshold.
- **Dynamic Adaptability:** Ensure high-fidelity voice extraction in unpredictable, non-stationary noise environments.

System Architecture: Operational Flowgraph



Experimental Setup & Expected Results

Current Focus: Software

- **Core Framework:** Python & TensorFlow for CNN training.
- **Spatial Modeling:** MATLAB for 3-mic TDOA & beamforming.
- **Datasets:** Microsoft DNS & UrbanSound8K for training.
- **Validation Goal:** Prove algorithm and spatial logic virtually first.

Future

Next Step: Hardware

- **Target Platform:** Xilinx Zynq / Intel FPGA.
- **Sensors:** $3\times$ I2S MEMS mic linear array.
- **Translation:** VHDL/Verilog for RTL mapping.
- **Optimization:** INT8 Quantization for footprint reduction.

Target Performance Metrics:

- **Signal-to-Noise Ratio (SNR) Improvement:** > 12 **dB**
- **Total Processing Latency (AI + DSP):** < 10 **ms**

Project Roadmap & Phased Execution

- **Phase 1: Research & Architecture Design (Completed)**
  - Studied the evolution of ANC from traditional FxLMS to modern Edge AI.
  - Finalized a hybrid architecture combining beamforming with neural network classification.
- **Phase 2: Software Modeling & Simulation (Next Steps)**
  - Train the lightweight CNN on standard datasets (UrbanSound8K) using TensorFlow.
  - Simulate the 3-mic TDOA spatial filtering in MATLAB to prove the concept in software.
- **Phase 3: Hardware Implementation**
  - Translate software models into hardware description languages (Verilog/VHDL).
  - Map the CNN Multiply-Accumulate (MAC) units to the physical FPGA development board.
- **Phase 4: Testing & Optimization**
  - Apply INT8 quantization to minimize FPGA logic footprint and power consumption.
  - Benchmark physical latency to ensure the system achieves the sub-10ms target.

Conclusion and Future Roadmap

Summary of Research

This research successfully bridges the gap between high-speed DSP and intelligent AI denoising. By implementing a **dual-mode architecture**, we have addressed the "Intelligence-Speed Paradox," providing a system that is both context-aware and fast enough for real-time human interaction. The integration of spatial discrimination ensures that the system doesn't just "suppress noise" but actively "extracts intelligence" from complex acoustic environments.

Key Contributions & Future Work

- **Architectural Efficiency:** A streamlined pipeline achieving < 10ms latency suitable for FPGA/Embedded deployment.
- **Context-Aware Switching:** Robust VAD logic that prevents speech distortion while maximizing noise attenuation.
- **Future Enhancement:** Integration of *Binaural Rendering* to provide a 3D spatial audio experience for the user.
- **Hardware Scaling:** Migrating the CNN core to dedicated NPU (Neural Processing Unit) hardware for even lower power consumption.

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